

Microwave Frequency Bands

Frequency Band	Frequency Range	Applications
L Band	1 to 2 GHz	❖ GSM (900MHz), CDMA (800MHz and 1.9GHz) and Satellite Phone
S Band	2 to 4 GHz	❖ Bluetooth (2.4GHz), Wi-Fi (2.4GHz), ZigBee (2.4GHz) and Weather & Surface Ship RADAR.
C Band	4 to 8 GHz	❖ Satellite Communication and RADAR
X Band	8 to 12 GHz	❖ Education, Satellite Communication, RADAR and Remote Sensing
Ku Band	12 to 18 GHz	❖ VSAT (Voice and Data) and DTH Satellite TV
K Band	18 to 27 GHz	❖ RADAR and Infrared Astronomy (Analysis of Planets and Stars)
Ka Band	27 to 40 GHz	❖ Low Range High-Resolution RADAR and Military Airplane
V Band	40 to 75 GHz	❖ High Capacity Terrestrial mm-Wave Communication (High Data Rate)
W Band	75 to 110 GHz	❖ mm-Wave RADAR and Research
mm Band	110 to 300 GHz	❖ mm-Wave RADAR and Satellite Communication

Advantages of Microwave

- ❑ It has a Large Bandwidth {1GHz-300GHz}, Hence Microwave has higher data rate.
- ❑ Highly Directive Properties of Microwaves, Due to that Microwaves can be used for Line of Sight (LOS) Communication {Satellite Communication}.
- ❑ Circuit Size is Very Small at Microwave {Circuit Size $\propto$ Wavelength λ }.
- ❑ Microwave is used for Long Distance Communication, It can penetrate through Fog, Rain, Clouds, etc.
- ❑ Microwave has an Efficient Size of Antenna {Antenna Size $\propto$ Wavelength λ }.
- ❑ Fading is very Less with Microwave, Multipath Fading Less with Microwave.
- ❑ Beamwidth is very Narrow with Microwave {For Reflector Antenna Beamwidth = $70\lambda/D$ }.

$$\leadsto f_1 = 300 \text{ MHz}$$

$$\lambda_1 = c/f_1 = \frac{3 \times 10^8}{300 \times 10^6} = 1 \text{ m}$$

$$\theta_1 = \frac{70 \times \lambda_1}{D} = \frac{70 \times 1}{1} = 70^\circ$$

$$\leadsto f_2 = 30 \text{ GHz}$$

$$\lambda_2 = c/f_2 = \frac{3 \times 10^8}{30 \times 10^9} = 0.01 \text{ m}$$

$$\theta_2 = \frac{70 \times \lambda_2}{D} = \frac{70 \times 0.01}{1} = 0.7^\circ$$